REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Bohr's Model of the Atom Practice Questions

Grade 12 Advanced | Module 22 Lesson 3 | Bohr's Model of the Atom

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Hydrogen: $E_n = -13.6/n^2$ eV; $r_n = (5.29 \times 10^{-11})n^2$ m; $E_{photon} = |E_f - E_i|$; $\lambda(nm) \approx 1240/E(eV)$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 62–74.

P01. Rutherford's large-angle alpha-particle scattering implied a nucleus that is:

A. Spread uniformly through the entire atom B. Negatively charged and larger than the atom
C. Without mass D. Small dense and positively charged

P02. In Bohr's model, permitted electron energies are:

A. Discrete B. Continuous at every radius
C. Always positive D. Independent of principal quantum number

---

P03. A photon is emitted when an electron:

A. Moves from lower to higher energy without external input B. Moves from higher to lower energy
C. Stays in the same stationary state D. Disappears without energy transfer

---

P04. Negative hydrogen bound-state energy means the electron:

A. Has negative mass B. Cannot absorb a photon
C. Needs added energy to reach the zero-energy free state D. Has zero binding energy

---

P05. The Balmer series of hydrogen ends at:

A. n = 1 B. n = 3
C. n = infinity D. n = 2

P06. Calculate the hydrogen energy level E_n for n = 3, using $E_n = -13.6/n^2$ eV.

A. -3.022 eV B. -0.7556 eV
C. -1.511 eV D. 1.511 eV

P07. Find the hydrogen orbital radius for n = 3. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

A. 9.52×10^{-10} m B. 2.38×10^{-10} m
C. -4.76×10^{-10} m D. 4.76×10^{-10} m

P08. A hydrogen electron falls from n = 4 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

A. 2.55 eV B. 5.1 eV
C. 1.275 eV D. -2.55 eV

P09. Find the minimum energy needed to ionise hydrogen from n = 3, using $E_n = -13.6/n^2$ eV.

A. 3.022 eV B. 1.511 eV
C. 0.7556 eV D. -1.511 eV

P10. An atomic transition emits a photon of energy 2.5 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

A. 992 nm B. 248 nm
C. 496 nm D. -496 nm

P11. Distinguish ground state from excited state.

P12. Why are line spectra characteristic of an element?

P13. What major problem did the classical planetary atom model face?

P14. What is the zero-energy reference in the simple hydrogen model?

P15. State one limitation of the Bohr model.

P16. Calculate the hydrogen energy level E_n for n = 4, using $E_n = -13.6/n^2$ eV.

P17. Calculate the hydrogen energy level E_n for n = 5, using $E_n = -13.6/n^2$ eV.

P18. Find the hydrogen orbital radius for n = 4. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

P19. Find the hydrogen orbital radius for n = 5. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

P20. A hydrogen electron falls from n = 5 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

P21. A hydrogen electron falls from n = 6 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

P22. Find the minimum energy needed to ionise hydrogen from n = 4, using $E_n = -13.6/n^2$ eV.

P23. Find the minimum energy needed to ionise hydrogen from n = 5, using $E_n = -13.6/n^2$ eV.

P24. An atomic transition emits a photon of energy 3 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

P25. An atomic transition emits a photon of energy 3.5 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

P26. For hydrogen, calculate the photon energy emitted for the n = 3 to n = 2 transition.

P27. Why do many alpha particles pass through thin gold foil with little deflection while a few scatter strongly?

P28. Distinguish emission and absorption line spectra.

P29. A photon energy is smaller than every available upward transition from a hydrogen atom's ground state. Can it excite that atom through a single bound-state transition?

P30. Compare hydrogen orbital radius and binding-energy magnitude as n increases.